EUROPEAN COMMISSION 2026

Claude Code for Empirical Research

Agents, workflows, and difference-in-differences

Scott Cunningham · Baylor University

European Commission · June 3–4, 2026

Automating empirical research is no longer hypothetical

- **Natural sciences:** AlphaFold predicted protein structure at scale. Nobel Prize in Chemistry, 2024.
- **Social sciences:** Stanford's *Social Catalyst Lab* is automating end-to-end empirical pipelines — data, design, estimation, write-up.
- **The question is not whether. It's how much H you intentionally keep.**

An agent is an LLM wrapped in a loop

LLM alone	One prompt. One reply. Done.
Agent	Same LLM, but it can use tools, observe results, and decide what to do next — repeatedly.

Every small decision inside that loop is recorded in the session log.
We can read them.

Chatbot vs. Copilot vs. Agent

	What it does	Who steers
Chatbot	Answers your question	You, every turn
Copilot	Completes what you’re typing	You, inline
Agent	Plans, acts, iterates, and verifies	You set the goal

Today: agents. Claude Code is an agent.

Research output is multiplicative

$$P = F_1 \times F_2 \times \cdots \times F_8$$

Shockley (1957) · *On the Statistics of Individual Variations of Productivity in Research Laboratories*

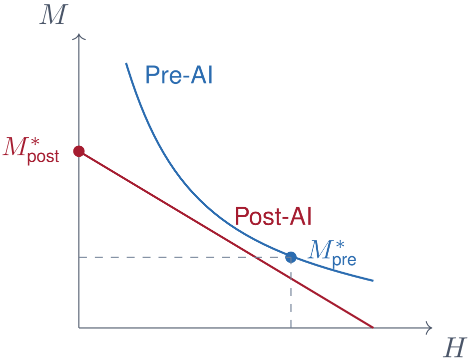
Any factor at zero kills output entirely.

Shockley's eight factors

1. Ability to think of a good problem
2. Ability to work on it
3. Ability to recognize a worthwhile result
4. Ability to decide when to stop and write up
5. Ability to write adequately
6. Ability to profit constructively from criticism
7. Determination to submit the paper to a journal
8. Persistence in making changes after journal action

Proxied by: publications, grants, patents.

AI flattens the isoquants of cognitive production

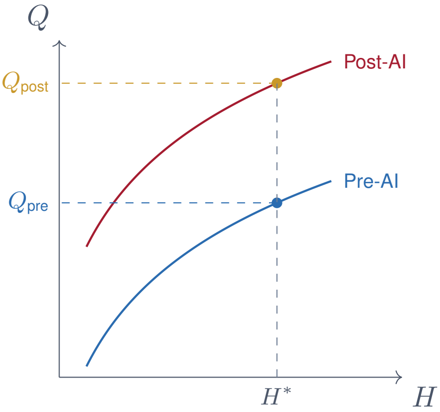


$$Q = f(H, M)$$

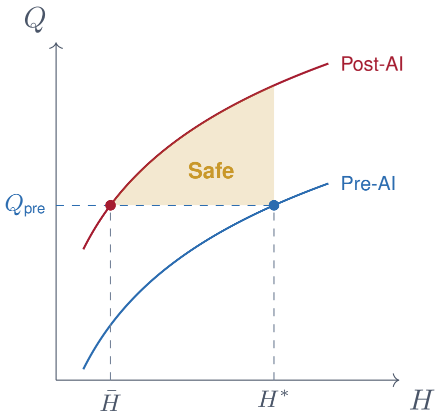
- H = human time and attention
- M = machine time

The question isn't whether to use M.
It's how much H you keep.

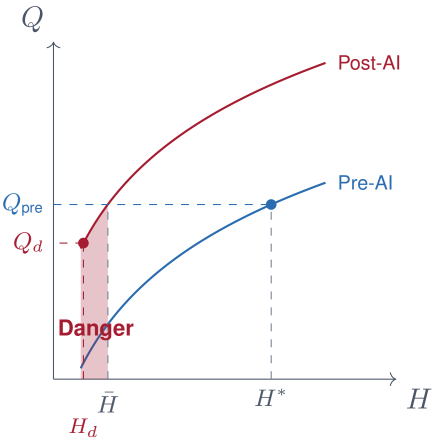
AI shifts the production curve up



Safe: take time back, stay above $\bar{H}$



Danger: $H < \bar{H}$, skills depreciate



Download Claude Code from the official source only

`claude.com/download`

Start with the **desktop app** — no terminal required to begin.

Available for macOS and Windows.

A new phishing attack: fake install sites

Fake “Claude Code install” pages are circulating. They ask you to copy and paste a command into your terminal. That command installs malware.

- The threat won’t come through email
- It looks like a legitimate software install page
- **Never paste terminal commands from a website you didn’t navigate to yourself**

Source: TechRepublic (2026) · `claude.com/download` is the only safe path.

MixtapeTools: skills for empirical research

`github.com/scunning1975/MixtapeTools`

A library of Claude Code skills built for researchers.

`/newproject` `/split-pdf` `/referee2`

`/beautiful_deck` `/blindspot` `/compiledeck`

`/newproject`: scaffold a project in seconds

```
/newproject
```

Creates:

- Standard directory structure (`data/`, `code/`, `output/`, `docs/`)
- `CLAUDE.md` with project context
- `README.md`

Give Claude access to your folder and a chapter from the Mixtape as context.

Starting a zero-shot now — in the background

One prompt. No instructions. Just a research question.

Objective: develop a DiD study using Italian or European data.

Binary treatment. Staggered adoption.

CS estimator.

We'll come back to this at the end of the day and see what was produced.

AER is already seeing the wave

Friend 1 · 10:17 PM

"Just learn that AER is already seeing a big inflow of AI papers lol"

"Written by AI. They are very concerned."

"No way they can handle 20–50% more papers"

"12 editors each handling 200 papers right now"

Friend 2 · 10:18 PM

"Haha written by AI or about AI?"

"$1,000 submission fee!"

This conversation happened this week.

Zurich Social Catalyst Lab: papers from the workshop

After the Zurich workshop (May 2026):

Participants built working papers using Claude Code and the methods from the course.

Zero-shot papers are not the ceiling. They are the floor.

We’ll do this together — live

One DiD paper, built in plan mode, using CS.

Italian or European data. Staggered adoption. We know the estimator.

It will run **all day**.

We’ll come back at the end and see where it is.

The workflow: nine steps in plan mode

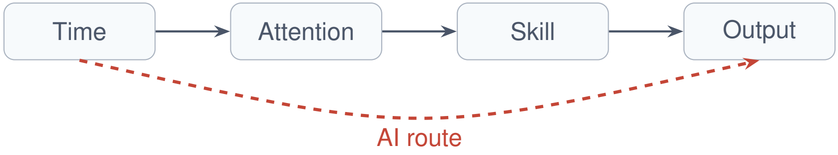
1. **Choose covariates** via interview process
2. **Check balance** with balance tables
3. **Build propensity score**
4. **Visualize rollout** with `panelview`
5. **Rollout graphic** with cohort counts and sample shares
6. **Outcome by cohort** over time
7. **Choose target parameter** (weighted ATT)
8. **Estimate** simple average + event studies with CS
9. **Sensitivity analysis** with Rambachan–Roth

Verification: the human stays in the loop

`/referee2`	Results in a second language / framework
GTD harness	Organize and track multi-step pipelines
`/beautiful_deck`	Compile results into a presentation

The agent does the work. You verify the design and the findings.

The traditional path from time to output



The AI route shortcircuits Attention and Skill.
Output arrives faster — but the pathway that builds the researcher is skipped.

Production costs have fallen

AER is already seeing a 20–50% inflow of AI-written submissions.

12 editors. Each handling 200 papers. Right now.

Submissions are rising — verification is not

- Agents reliably produce output (1): code ran, estimates exist
- They do not guarantee correct output (2): right estimator, right design
- They do not guarantee interpretable output (3): can you reconstruct every step?

Production was hard; verification was implicit.
Now production is cheap; verification is hard.

You will need to build a verification workflow

Verification is the new bottleneck.

- `/referee2` — replicate results in a second language or framework
- `/blindspot` — surface what the analysis missed
- `/beautiful_deck` — force the agent to present its own findings

The workflow doesn't exist yet. You will probably need to develop one for yourself.

The agent does the work.

You set the question.

You verify the design.

You own the answer.